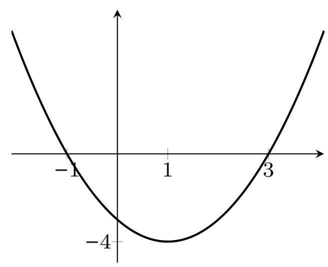
Problem Set 29

# The Fundamental Theorem of Calculus, Part 1

**1** Let $f$ be the function whose graph is shown below, and let $F(x) = \int_2^x f(t)\,dt$.




(a) Evaluate $F(2)$.

**Solution**

$F(2) = \int_2^2 f(t)\,dt = 0.$

(b) Is $F(3)$ positive, negative, or zero? Explain how you know.

**Solution**

$F(3) = \int_2^3 f(t)\,dt$, which we can visualize as the signed area under $f(t)$ from $t = 2$ to $t = 3$. This signed area is $\boxed{\text{negative}}$ since $f < 0$ on $(2,3)$.

(c) Is $F(1)$ positive, negative, or zero? Explain how you know.

**Solution**

$F(1) = \int_2^1 f(t)\,dt = -\int_1^2 f(t)\,dt$. The integral $\int_1^2 f(t)\,dt$ can be visualized as the signed area under $f(t)$ from $t = 1$ to $t = 2$, and this signed area is negative. Therefore, $F(1)$ is $\boxed{\text{positive}}$.

(d) For what values of $x$ is $F(x)$ increasing?

**Solution**

$F$ is increasing when $F'$ is positive; by FTOC 1, $F' = f$, and this is positive on $\boxed{(-\infty,-1) \text{ and } (3,\infty)}$.

(e) For what values of $x$ is $F(x)$ concave up?

**Solution**

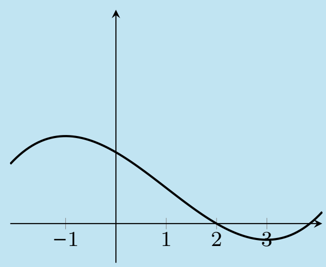
$F$ is concave up when $F''$ is positive; by FTOC 1, $F'' = f'$, which is positive when $f$ is increasing. This happens on $\boxed{(1, \infty)}$.

(f) Sketch a rough graph of $F(x)$ on the interval $[-2, 4]$.

*Make sure it agrees with everything you've said above!*

**Solution**

We've seen that $F$ has an $x$-intercept at $x = 2$, and it's increasing on $(-\infty, -1)$, decreasing on $(-1, 3)$, and increasing on $(3, \infty)$. Also, it's concave down on $(-\infty, 1)$ and concave up on $(1, \infty)$. Thus, its graph looks like this:



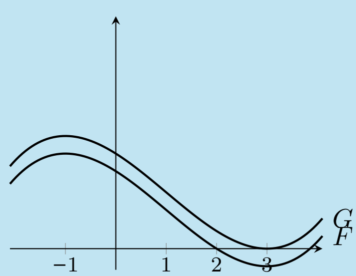

(g) Now, let $G(x) = \int_3^x f(t)\,dt$. Sketch a rough graph of $G(x)$. (*Hint:* What do you know about how $G(x)$ relates to $F(x)$? What is $G(3)$?)

**Solution**

Since $G$ and $F$ have the same derivative (both $G'$ and $F'$ are equal to $f$ by FTOC 1), they are vertical shifts of each other. Another way to see that $G$ and $F$ are vertical shifts of each other is to write

$$F(x) = \int_2^x f(t)\,dt = \int_2^3 f(t)\,dt + \int_3^x f(t)\,dt = \underbrace{\int_2^3 f(t)\,dt}_{\text{a constant}} + G(x);$$

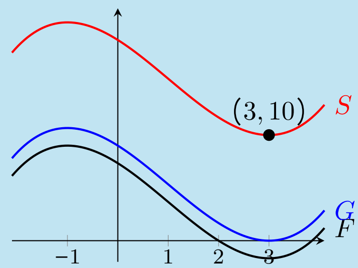
We also know that $G(3) = 0$, so we shift the graph of $F$ vertically so that it crosses the $x$-axis at $x = 3$:




(h) Suppose that $f(t)$ represents the rate, in students per minute, at which students are entering or leaving Annenberg $t$ minutes after 8 am one morning. If there are 10 students in Annenberg at 8:03 am, sketch the graph of $S(t)$, the number of students in Annenberg $t$ minutes after 8 am.

**Solution**

At $t = 3$, there are 10 students in Annenberg. Therefore, the number of students in Annenberg $t$ minutes after 8 am is $10 + \int_3^t f(u)\,du$. That is, $S(t) = 10 + G(t)$. Therefore, to graph $S(t)$, we just shift the graph of $G(t)$ up by 10 units:




**2** Let $f(x) = \dfrac{2x}{x^2+1}$. Laura is trying to find a formula for the area function $F(x) = \int_1^x f(t)\,dt$.

(a) What is $F(1)$?

**Solution**

$F(1) = \int_1^1 f(t)\,dt = \boxed{0}$.

(b) What is $F'(x)$?

**Solution**

By FTOC 1, $F'(x) = f(x) = \boxed{\dfrac{2x}{x^2+1}}$.

(c) One of the following is a formula for $F(x)$; which one? (*Note:* You are **not** expected to be able to directly find a formula for $F(x)$; however, given what you found in the previous parts of this problem, you should be able to eliminate all but one of the choices below.)

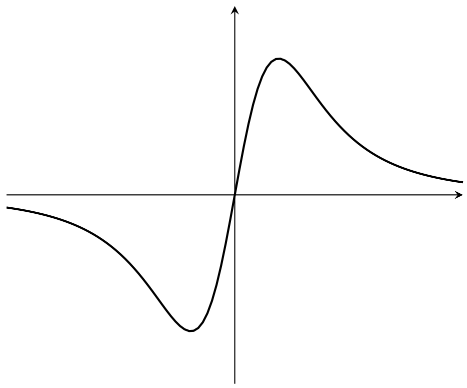
A. $\ln(x^2+1)$ B. $\ln(x^2+1) - \ln 2$ C. $\dfrac{2(1-x^2)}{(x^2+1)^2}$ D. $2x \arctan x$

Explain your reasoning.

**Solution**

We know two things about $F(x)$: its derivative is $\dfrac{2x}{x^2+1}$, and it has an $x$-intercept at $x = 1$. Only (A) and (B) have derivative $\dfrac{2x}{x^2+1}$; of those two, only $\boxed{\text{(B)}}$ has an $x$-intercept at $x = 1$.

**3** Here is the graph of a function $f(t)$.



(a) One of the graphs below (choices A–K) is the graph of $F(x) = \int_0^x f(t)\,dt$; which one? Explain your reasoning.

**Solution**

By FTOC 1, $F'(x) = f(x)$. In particular, since $f$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$, $F$ is decreasing on $(-\infty, 0)$ and increasing on $(0, \infty)$. This only matches (D) and (E).

Since $F(0) = \int_0^0 f(t)\,dt = 0$, the graph of $F$ must be $\boxed{\text{(D)}}$.

(b) One of the graphs below is the graph of $G(x) = \int_2^x f(t)\,dt$; which one? Explain.

**Solution**

By FTOC 1, $G'(x) = f(x)$; since $F$ and $G$ have the same derivative, they must be vertical shifts of each other. The only vertical shift of $F$ is $\boxed{\text{(E)}}$.

(Also, notice that $G(2) = \int_2^2 f(t)\,dt = 0$.)

(c) One of the graphs below is the graph of $H(x) = \int_x^2 f(t)\,dt$; which one? Explain.

**Solution**

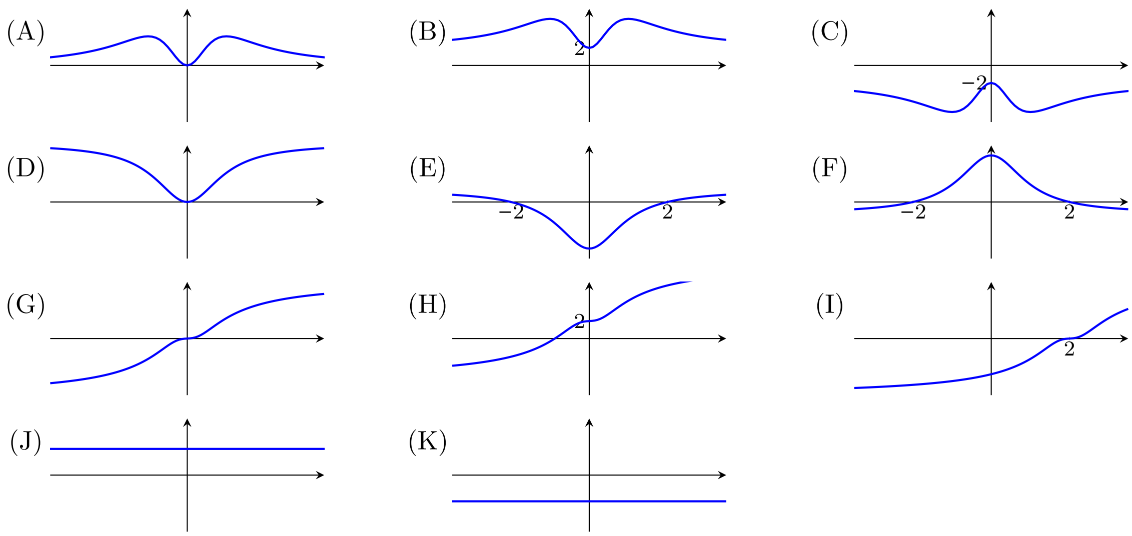
We can write $H(x) = \int_x^2 f(t)\,dt = -\int_2^x f(t)\,dt = -G(x)$. So, to get the graph of $H$, we can flip the graph of $G$ over the $x$-axis, which gives us graph $\boxed{\text{(F)}}$.

(d) One of the graphs below is the graph of $K(x) = \int_2^3 f(t)\,dt$; which one? Explain.

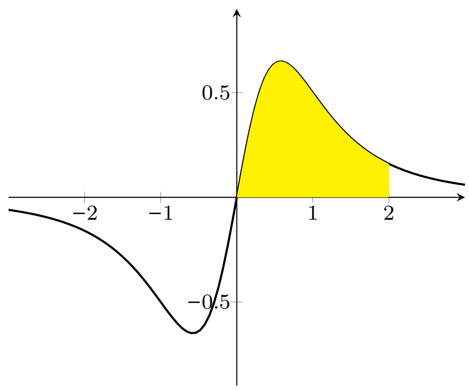
**Solution**

$K(x)$ is a positive constant, so its graph is $\boxed{\text{(J)}}$.

Choices:




**4** This problem uses the function $f(t)$ from #3, graphed again below.




Define the functions $F(x) = \int_0^x f(t)\,dt$ and $G(x) = \int_2^x f(t)\,dt$.

(a) Calculate $F'(0)$ and estimate $F'(1)$.

**Solution**

$F'(0) = f(0) = \boxed{0}$ and $F'(1) = f(1) \approx \boxed{0.5}$.

(b) Calculate $G'(0)$ and estimate $G'(1)$.

**Solution**

$G'(0) = f(0) = \boxed{0}$ and $G'(1) = f(1) \approx \boxed{0.5}$.

(c) If we define $J(x) = G(x) - F(x)$, what is $J'(0)$? What is $J'(1)$?

**Solution**

Since $J(x) = G(x) - F(x)$, $J'(x) = G'(x) - F'(x)$ for any $x$. So

$$J'(0) = G'(0) - F'(0) = \boxed{0}$$

and

$$J'(1) = G'(1) - F'(1) = \boxed{0}.$$

(d) On the graph of $y = f(x)$, shade the region whose area is equal to $|J(0)|$.

**Solution**

$J(0) = \int_2^x f(t)\,dt - \int_0^x f(t)\,dt = \int_2^0 f(t)\,dt = -\int_0^2 f(t)\,dt$. So $|J(0)|$ is the area under the curve between $x = 0$ and $x = 2$ (shaded in yellow on the graph at the start of the problem); and $J(0)$ is the negative of this.

(e) Looking at the graphs (A)–(K) in problem #3: which of them could be the graph of $y = J(x)$? Explain your answer fully.

**Solution**

Since $G'(x) = F'(x)$ for all $x$, we can see that $J'(x) = 0$ for all $x$. This means that $J$ is never increasing and never decreasing, which means our answer is either (J) or (K). From (d), we see that $J(x)$ is always negative. Therefore, the answer is $\boxed{\text{(K)}}$.